

Citation under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

Harsh DUBEY¹

New York University, New York, USA

Abstract. Sanctions over fabricated legal citations have made language-model drafting a problem for courts, and most research on the problem has produced detection benchmarks that score errors in finished briefs. This paper measures how the errors are produced. Seven models from seven vendors, from a 30B open-weight model to frontier systems, drafted argument sections for 48 U.S. Supreme Court matters under five prompting conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 drafts was checked deterministically, with no language-model judge in any primary measurement: existence against an 18-million-row reporter index, quotations against a verbatim checker validated on seeded faults, and pinpoint pages against official reporter pagination from public archives. Under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, and verbatim quotation accuracy fell for six of seven models, a fall that survives correction once three generations or two prompt templates are pooled; Sonnet 5 was the exception. In a second experiment the checker fed its findings back into revision. Sonnet 5 reached 0.974 verbatim accuracy, and a per-quotation trace shows why. Of its 89 flagged quotations, 4 were corrected; the rest were deleted or stripped of their quotation marks. False feedback produced the same deletions aimed at correct quotations, five of seven models ended less accurate than they started, and a verifier right half the time cost Sonnet 5 more than twice the correct quotations a precise one did. We release the instruments, including a page-level pinpoint verifier built from public data.

Keywords. legal citation, hallucination, large language models, verification, evaluation

1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs containing citations that a language model invented, beginning with *Mata v. Avianca* [1] and continuing in a growing docket that a public tracker now counts in the hundreds [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model or agent find [3,4]? Detection accuracy is the right measure for re-

¹Corresponding Author: Harsh Dubey, New York University; hd2225@nyu.edu.

viewing a filed brief; it does not tell a deployer which prompting conditions produce the errors, which is the question this paper addresses.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and does the answer depend on the model? Deployed prompts can carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a minimum count of authorities, a restriction on which may be cited, and a warning about consequences collapsed citation existence even in capable models; we are not aware of a comparable study for law. Second, when a checker that can verify citations deterministically feeds its findings back into revision, does the model correct its drafts, fail to, or change the draft so that the checker’s tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the best published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall), and its authors conclude that “no models can reliably detect wrong pincites” because page-level information lives behind commercial databases [3]. We built it from public sources: existence and name pairing against a local index derived from CourtListener bulk data, verbatim quotation checking validated on seeded faults, and page-level pinpoint verification against the official reporter pagination in the Caselaw Access Project’s volume archives [6]. Every primary number reported here is produced by this instrument; no language-model judge is used in any primary measurement.

Our primary setting is the closed-book use of general-purpose models, the setting of most sanctioned filings documented to date [2,7]; a grounded arm, in which each prompt carries ten retrieved and verified authorities, measures what retrieval changes. The paper reports three results. First, pressure degrades citation integrity beginning with the smallest model, and the degradation is rarely accompanied by refusal; in the 1,344 drafts under the four non-baseline conditions (the pressured drafts) there were three refusals. Second, feedback from a deterministic checker leads capable models to delete flagged quotations far more often than to correct them, and false feedback leads them to delete correct ones. Third, at the top of the range the remaining errors are mostly paraphrase presented as verbatim quotation, genuine language from a different case, or the wrong page, which the released instruments detect.

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their error categories as outcome coding and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book for the citation of a named case almost never produce it and almost never abstain; our task asks the easier question of whether the cases a model chooses to cite exist. Dahl et al. [7] measured hallucinated authority in general-purpose chatbots and Magesh et al. [8] in commercial legal research tools.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified by fuzzy bibliographic matching that left 36 to 61 percent of citations unresolved. We transplant their factorial to law with deterministic verification (0.2 percent unresolved) and a formal statistical model; their mechanism, compliance with the temporal window filled with fabrications, transfers to law for some models only (Section 4).

Revision with execution feedback. The loop protocol follows execution-feedback work in code generation [9], including a no-feedback control that separates feedback content from extra revision attempts; we add controls in which half or all of the feedback is false.

AI and law. Blair-Stanek and Van Durme [10] found at ICAIL that frontier models flip the winning party across identical zero-temperature runs; the primary run generated each cell once, and Section 8 treats that instability as a limitation of this design. Evaluation methodology in this community has moved toward separating factual truth from reasoning soundness [11] and pairing automated metrics with the checks lawyers perform [12]; citation integrity is a check for which deterministic ground truth is available, and this paper automates it. Closest to our pinpoint instrument, Garneau et al. [13] found that retrieving the paragraph a proposition rests on is far harder than retrieving the case; our verifier approaches the same gap from the other side, checking a generated pincite against the page text. Zhang et al. [14] show models miss overruled precedent, a failure adjacent to the wrong-authority errors we measure, and ConsRAG [15] constrains retrieval to prevent hallucination, the strategy complementary to the post-hoc feedback our loop tests. On trust in model-drafted documents, Trozze et al. [16] found mock jurors indifferent to whether a complaint was written by ChatGPT or a lawyer, and Harasta et al. [17] found lawyers rate identical documents lower when told a model wrote them; machine verification offers a basis for trust that depends on neither perception. Gray et al. [18] and Pires et al. [19] evaluate generated legal argument without checking the authority it cites, the layer this paper adds; citation-worthiness detection [20] and citation-network analysis [21,22,23] study the same network our checker uses to test whether a cited case exists.

3. Tasks, Conditions, and Instruments

Terms. In a citation such as *Brown v. Board of Education*, 347 U.S. 483, 495 (1954), the volume, reporter, and first page identify the case and the optional second page (495) is the *pincite*, which for a quotation must be the page where the quoted language appears. Throughout, *citation existence* is the share of adjudicable citations that resolve to a real case, and *strict quotation accuracy* is the share of scored quotations that reproduce the opinion verbatim.

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020, drawn from a 500-case pool built from the Supreme Court Database [24]. Each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, with the Supreme Court’s decision excluded and every packet screened for outcome leakage. The model argues an assigned side (petitioner for even-numbered matters, respondent for odd) in an argument section of under 900 words; nothing in the design predicts outcomes, so the memorization confound of Supreme Court prediction tasks [25] is out of scope.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite **MUST** have been decided before 1970”); stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which concatenates the three clauses. Each condition has one fixed template, run at temperature

0. The stakes clause names Rule 11 of the Federal Rules of Civil Procedure because it is the sanctions rule invoked in the sanctioned-filing cases [1]; Rule 11 governs civil actions in the district courts [26], so the clause serves only as a stakes signal for a Supreme Court filing. Each condition is a single clause so that its effect can be isolated. A second template, an alternative phrasing of every condition, was run on all 48 matters and seven models. A *grounded* arm reran baseline and combined with ten retrieved authorities in the prompt: the U.S. Reports opinions most similar to the question and facts (TF-IDF over the first 2,500 words of the 25,250 opinions in our Caselaw Access Project cache, decided before argument and before 1970 for the combined condition, any opinion naming either party excluded), each with citation, year, and its best-matching 250-word passage, stated to be real, with quotation restricted to the excerpts and other authority permitted.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, all accessed through one API gateway. The set spans open-weight 30B to frontier with no vendor contributing two models; we do not assume a total ordering, tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are called the top of the range. The gateway identifiers, run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens, are qwen3-30b-a3b-instruct-2507, mistral-small-2603, llama-4-maverick, deepseek-v4-flash, grok-4.3, gpt-5.4-mini, and claude-sonnet-5. An eighth model, GLM-4.7-flash, completed 207 of 240 drafts and is reported in Section 4.

Adjudication. Citations are extracted with eyecite [27] and given one of three labels: exists, not found, or unverifiable (vendor citations and coverage gaps), so that index coverage is never counted as fabrication; 0.2 percent of the 14,040 citation records in the full run are unverifiable. Existence and name pairing resolve against an 18-million-row citation table; quotations are located by balanced quotation marks and attributed to the case named in the same sentence, else to the nearest citation within 260 characters after it, else to the nearest preceding one. Each is checked verbatim against the cited opinion’s text from public sources; a quotation is *near miss* when every fragment reaches token coverage of at least 0.85 without matching exactly, and *inaccurate* otherwise. The strict rate for a draft d is $S_d = n_{\text{acc}} / (n_{\text{acc}} + n_{\text{near}} + n_{\text{inacc}})$, and a lenient rate counting near misses as correct is reported alongside it. Before any measurement the quotation checker scored 100 of 100 on twenty synthetic drafts with planted genuine quotations, wrong attributions, single-word corruptions, fabrications, and formatting artifacts; Section 6 audits its verdicts on real drafts. Pinpoint pages are verified against the official reporter pagination parsed from the Caselaw Access Project’s HTML volumes, which mark every page break; a quotation counts as on the cited page when its longest fragment lies within that page’s text, and adjacent-page matches are reported separately.

Human reference. The same instrument scored 482 clean, pre-ChatGPT federal appellate brief excerpts from Liu et al. [3]; human lawyers reach 0.407 strict and 0.680 lenient quotation accuracy (578 scored quotations) and 0.931 citation existence (the shortfall is index coverage of state reporters). Human briefs cite more non-federal authority and quote statutes and records more often than the models, so the human figures are a reference for the severity of the verbatim standard only.

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$ for citation i in matter m under condition k , with exchangeable working correlation within matters [28], so that the 48 matters

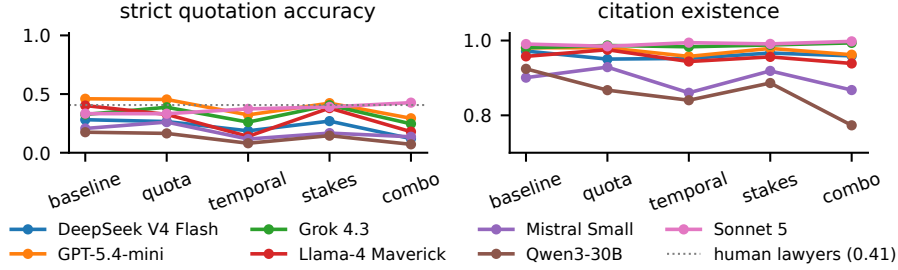


Figure 1. Strict quotation accuracy (left) and citation existence (right) by condition, one line per model. The dotted line is the human-lawyer reference under the identical instrument. The right panel’s axis starts at 0.7.

Table 1. Baseline versus the combined condition, rows ordered by baseline strict quotation accuracy. Quotation columns give the strict rate with the lenient rate in parentheses; existence columns give the rate with the raw count of citations not found in brackets. An asterisk marks a contrast significant after Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Temporal viol.
Qwen3-30B	0.175 (0.564)	0.072 (0.322)*	0.924 [28]	0.773 [90]*	20%
Mistral Small	0.207 (0.608)	0.136 (0.383)	0.901 [37]	0.868 [69]	35%
DeepSeek V4 Flash	0.281 (0.631)	0.120 (0.439)*	0.972 [10]	0.959 [20]	11%
Grok 4.3	0.326 (0.644)	0.246 (0.578)	0.981 [5]	0.994 [3]	0%
Sonnet 5	0.333 (0.622)	0.427 (0.642)	0.991 [3]	0.998 [1]	0%
Llama-4 Maverick	0.402 (0.618)	0.179 (0.453)*	0.958 [9]	0.939 [31]	10%
GPT-5.4-mini	0.460 (0.722)	0.294 (0.571)	0.981 [8]	0.962 [27]	13%

rather than the citations are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model and call an effect significant only when it survives correction, so statements that count models (“three of seven”) are descriptive. Odds ratios, 95 percent intervals, and raw and corrected p values for all 56 contrasts are in the released results, and a matter-level cluster bootstrap over draft rates agreed in sign with every corrected result. Two robustness fits follow the same model: one pools two further generations of the baseline and combined drafts (Section 8) with a run effect, Holm-corrected over its 14 tests; the other pools both templates with a template effect, corrected within model as above. For the revision experiment, which was pre-registered descriptively, we add matter-paired comparisons of final strict accuracy between arms (Wilcoxon signed-rank test and a bootstrap interval on the mean difference), chosen after the data were seen. Hypotheses and decision rules were frozen before the full run; every later analysis decision, including two withdrawn findings, is recorded in a dated ledger released with the code.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Citation existence. In the single run, only the 30B model’s existence moves under pressure: from 0.924 at baseline to 0.773 under the combined condition, with the quota,

temporal, and combined contrasts all significant after correction (combined OR 0.30, 95 percent CI 0.18 to 0.51, corrected $p < 0.001$). Mistral Small falls under the temporal and combined conditions without reaching significance, and the five other models stay between 0.93 and 1.00 in every condition.

Quotation accuracy. The temporal restriction is the condition that damages quoting. After correction, strict quotation accuracy falls significantly under temporal and combined pressure for Qwen3-30B (0.175 to 0.072) and Llama-4 (0.402 to 0.179), and under the combined condition for DeepSeek (0.281 to 0.120); GPT-5.4-mini, Mistral Small, and Grok 4.3 fall in the same direction without surviving correction in the single run, and Sonnet 5 does not fall at all. Llama-4 shows the largest drop. At baseline it quotes at the human reference level, 0.402 against 0.407, and under the temporal restriction its accuracy drops to 0.139 (OR 0.20, CI 0.10 to 0.40, corrected $p < 0.001$); the lenient rates in Table 1 move with the strict rates for every model, so the result does not depend on how bracketed alterations are scored. With three generations pooled, the combined-condition fall survives correction for 6 of the seven models (OR 0.33 to 0.50), Sonnet 5 being the exception, and Mistral Small's existence fall also survives (OR 0.62, corrected $p=0.003$).

Robustness to phrasing. Under the second template, 12 contrasts survive correction, 5 of the 8 that survived under the first; pooling both templates, all 8 survive and the combined-condition fall in quotation accuracy survives for 6 of seven models (OR 0.32 to 0.46), again all but Sonnet 5 (pooled OR 1.39, corrected $p=0.122$). Existence moves only for Qwen3-30B (second template 0.882 to 0.692) and, pooled, for GPT-5.4-mini under the temporal clause (OR 0.46, corrected $p=0.028$). Cell rates differ between templates by up to 11 points of quotation accuracy and 19 of existence, so single-cell rates depend on the template while the within-model contrasts do not.

Grounded drafting. With ten verified authorities in the prompt, the models drew 51 to 83 percent of their citations from the list, citations not found fell from 341 in 5,856 closed-book to 74 in 5,605, Qwen3-30B's combined-condition existence went from 0.773 to 0.995 (OR 53.8, corrected $p<0.001$), and combined-condition existence was no lower than baseline for any model. Quotation accuracy rose in every combined cell and 6 of seven baseline cells (Qwen3-30B combined 0.072 to 0.368; Sonnet 5 baseline 0.333 to 0.525, corrected $p<0.001$), yet with the source passages in the prompt no cell exceeded 0.525, and 62 percent of Sonnet 5's 221 remaining inaccurate quotations are paraphrases of the cited case in quotation marks, a higher share than closed-book.

A pre-registered contrast that did not survive correction. Sonnet 5's strict quotation accuracy is numerically highest under the combined condition (0.427 against 0.333; uncorrected $p = 0.0294$, corrected $p = 0.235$). The stakes condition alone moves nothing after correction, and the temporal restriction shifts the cited cases toward older, better-memorized opinions (median decision year 1943 against 1986), an untested alternative to any reading in terms of more conservative citing. With three generations pooled the existence contrast reaches corrected $p=0.005$ on 7 citations not found in 975 against 5 in 1,324, and the quotation contrast $p=0.051$; the direction is stable across runs, the memorization alternative is untested, and we draw no conclusion.

Compliance with the temporal rule. Violations of the pre-1970 restriction (Table 1) run from 35 percent of adjudicable citations for Mistral Small to 1 in 803 for Sonnet 5 and 2 in 719 for Grok 4.3 across the temporal and combined conditions. Mistral Small violates the rule most while its other figures are among the two or three lowest, so compliance does not track the other outcomes in rank.

Table 2. Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft; removed means deleted, dequoted, or made inaccurate; strict accuracy round 0 to final; Clean counts episodes ending on a passing draft.

Model	n	Flagged at round 0			Accurate at round 0		Strict	Clean
		corrected	dequoted	deleted	n	removed		
Qwen3-30B	170	0	11	30	16	0	0.099 to 0.107	0
Mistral Small	130	0	12	39	15	2	0.208 to 0.264	9
DeepSeek V4 Flash	183	0	67	101	28	12	0.181 to 0.645	21
Grok 4.3	46	0	27	19	15	5	0.207 to 1.000	24
Sonnet 5	89	4	13	72	68	17	0.471 to 0.974	23
Llama-4 Maverick	53	0	9	28	11	5	0.176 to 0.349	21
GPT-5.4-mini	37	2	5	21	28	2	0.360 to 0.719	22

Refusals and one model that stalls. Across the 1,344 pressured drafts there were three refusals (two DeepSeek, one GPT-5.4-mini). GLM-4.7-flash produced no output on 33 of its 240 drafts, 28 of them in the combined condition, reasoning until a 30,000-token budget was exhausted; it is excluded from Table 1 and reported as a third failure mode beside fabrication and refusal.

5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop. Each round, the checker returned one line per failure (a citation that does not resolve; a quotation that does not appear verbatim in the cited opinion), the model revised, and the loop ended on a clean draft or after three revisions. A control with *false feedback* gave the same number of identically formatted lines pointed at items that were correct; *mixed* controls kept each line or replaced it with a false one at random, so that the verifier’s precision was about 0.25, 0.5, or 0.75; and a *no-feedback* control gave the same number of revision rounds with a generic instruction to revise and no list of failures. The loop used the first 24 matters in the pre-registered order (twelve per side); loop rates are means over drafts on 24 matters, so round-0 figures differ from the pooled citation-level rates over 48 matters in Table 1, and citation counts in this section include unverifiable citations. Every scored round-0 quotation was traced to the final draft as *corrected* (the same or an edited quotation on the same citation is now accurate), *dequoted* (its words remain without quotation marks), or *deleted*.

Satisfying the checker by deletion. Under true feedback the final strict rate ranges from 0.107 for Qwen3-30B to 1.000 for Grok 4.3 (Table 2), and the rate alone is consistent with repair; the trace shows that the rise comes from deletion. Sonnet 5 had 89 flagged quotations at round 0: 4 were corrected, 13 kept their words and lost their quotation marks, and 72 were deleted; 17 of its 68 accurate quotations were removed as well. Its scored quotations fell from 157 to 73 while its accurate ones went from 68 to 70. Grok 4.3 reached 1.000 with every episode clean by correcting none of its 46 flagged quotations and dequoting or deleting all of them. Citation existence, by contrast, was repaired: the loop existence rate rose from 0.739 to 0.822 for Qwen3-30B and from 0.930 to 1.000 for Llama-4 while citation counts fell by at most a tenth. At every capability tested, checker feedback led models to remove flagged quotations or convert them to un-

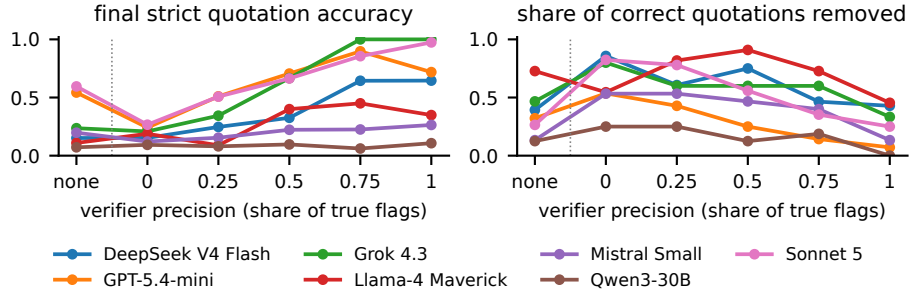


Figure 2. Verifier precision against what revision does. Left: final strict quotation accuracy. Right: share of the accurate round-0 quotations that revision deleted, dequoted, or made inaccurate. Precision is the share of feedback lines that were true; the leftmost point is revision with no feedback.

marked paraphrase far more often than to correct them; neither outcome is visible to the checker or to a reader of the revised draft.

Controls: feedback content and verifier precision. Figure 2 separates feedback content from revision itself and varies the verifier’s precision (per-arm counts are in the released data). Without feedback, Sonnet 5’s strict rate still rose (0.471 to 0.595), and true feedback added 0.36 over revision alone (matter-paired; interval 0.23 to 0.51, $p=0.0003$); the same holds for DeepSeek (0.49, $p=0.005$) and Llama-4 (0.29, $p=0.039$). For GPT-5.4-mini the difference between true feedback and no feedback is 0.12 with an interval that includes zero, so its apparent repair is not distinguishable from what revision alone does; Qwen3-30B moves little in any arm, and Mistral Small gains little from true feedback but falls to 0.122 under false feedback ($p=0.012$). The right panel of Figure 2 counts the correct quotations an imprecise verifier costs. For Sonnet 5 the correct quotations removed rise steadily as precision falls, 17, 24, 38, 53, and 56 of 68 at precisions of 1, 0.75, 0.5, 0.25, and 0, against 18 with no feedback, and final accuracy falls with them (0.974, 0.855, 0.664, 0.508, 0.266); a verifier right half the time left it within seven points of revision with no feedback (0.595). Under all-false feedback five of seven models ended below their starting rate (Llama-4 and Grok 4.3 within about a point of it), and inaccurate counts fell in every model, so what changed was the deletion of correct quotations. Our checker’s own false-alarm rate, from the provenance audit of Section 6, was 1.2 percent of Sonnet 5’s inaccurate verdicts (9 of 746).

6. Residual Errors at the Top of the Range

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion’s own text, 55 percent of Sonnet 5’s (413 of 746) are recognizable paraphrases of the correct case presented as verbatim quotation, 202 match nothing substantial in the cited opinion, and 131 cite authority outside the text store’s scope; GPT-5.4-mini’s split is 52 percent paraphrase.

Real language, wrong case. Searching every inaccurate quotation against a segmented corpus of 8,392 Supreme Court opinions locates 626 quotations across the seven models (127 for Sonnet 5, 77 for Qwen3-30B) whose language appears verbatim in a different opinion than the one cited (matched probes are at least six words, so stock legal

phrases do not count); quotations found nowhere in the corpus (511 for Sonnet 5) include the paraphrases of the preceding paragraph and quotations from other courts. The same search audits the checker: it classified 9 of Sonnet 5's inaccurate verdicts and 9 of Qwen3-30B's as the checker's own error, the language being present in the attributed case.

Right case, wrong page. Where a pinpoint and a quotation co-occur on a U.S. Reports authority, the quotation lies on the cited page 77 percent of the time for Sonnet 5, between 42 and 64 percent for the other six, against 89 percent for the human briefs of Section 3 (45 quotations), and on a different page of the cited case 16 to 45 percent of the time (the remainder are adjacent-page matches); pins outside the cited case are rare.

Relevance judging: three failed calibrations. Whether a real, correctly quoted case supports the proposition has no deterministic test; three language-model panels calibrated against the expert-labeled content-misrepresentation items of Liu et al. [3], with a pre-declared bar of 0.75 accuracy, scored 0.500, 0.700, and 0.594 (misrepresentation recall at best 4 of 17); no verdict from a failed panel was interpreted, and the result joins evidence that agreement statistics overstate judge validity [29,30].

7. Discussion

Where the checker belongs. The revision experiment argues for placing the checker at the gate, where it holds a draft and returns the failures to a person, and against placing it in the loop, where the model resolves them; the two differ only in who acts on the checker's output. Table 2 records Sonnet 5's revisions: 89 flagged quotations, 4 corrected, the rest removed. Removing an unverifiable quotation is a defensible response to a flag, and three facts keep it from being a repair. The words usually survive as paraphrase the checker cannot test and the reader cannot tell from a claim that was never quoted; false flags get the same response, so correct quotations go too (Figure 2); and a tool built on the loop pattern reports a clean draft, so the supervising attorney never sees the list. With the checker at the gate that list reaches the attorney.

Verifier precision is a deployment requirement. The mixed and false-feedback arms measure what an imprecise verifier costs. Our checker's false-alarm rate on Sonnet 5's drafts was 1.2 percent; published detection agents are less precise, and Liu et al. [3] report that weaker detectors flag a substantial share of correct citations, reading a coverage gap in a public index as a fabrication. Figure 2 shows what such flags do once a capable model acts on them: the correct quotations lost rise at every step as precision falls from 1 to 0, and at a precision of one half Sonnet 5 lost more than twice what it lost under a precise verifier. A pipeline that feeds a probabilistic detector's flags to a model should therefore report the detector's precision alongside its recall. For existence, verbatim quotation, and pinpoint page no probabilistic detector is needed; the check is exact and the released instrument performs it from public data.

What evaluations should include. A single fabrication rate at a neutral prompt misses both effects reported here: the 30B model's baseline existence of 0.924 overstates what it delivers under constraints a user is likely to add, and quotation accuracy, which existence benchmarks omit, fell for six of seven models, most under the temporal clause. The residual classes at the top of the range are invisible to existence checks, and the grounded arm shows that retrieval leaves them in place. With the source passage in the

prompt about half of the quotations at the top of the range are still not verbatim, and for every model most inaccurate ones are paraphrase in quotation marks.

For courts. Court orders that require disclosure of AI assistance [2] ask filers to state whether a model was used; the three checks reported here can be run by the court on any filing and cover the failures named in the sanctions decisions [1].

8. Limitations

The study covers one domain and task, Supreme Court merits argument in one packet format with two templates per condition, closed-book except for one retrieval arm with a single lexical retriever, and the stakes clause names a sanctions rule that does not govern the forum. The no-feedback and mixed controls, the grounded arm, and the per-quotation trace were added after the pre-registered analysis, and the tests in Section 5 were chosen after the data were seen. Each cell was generated once at temperature 0. Two further independent generations of the baseline and combined drafts on all seven models (672 drafts per run) produced no draft identical to another run's except 22 of Mistral Small's, the instability reported by Blair-Stanek and Van Durme [10]; across the three runs the standard deviation of a cell's existence rate was at most 0.050 and of its strict quotation rate at most 0.059, and the combined-versus-baseline quotation contrast kept its sign in every run for all 7 models. Single-run cells therefore carry noise of a few points, and the pooled three-run fit reported in Section 4 is the more precise estimate of the combined-condition effect. Two templates per condition do not span the phrasings users write; the quotation rates in Table 1 are template-specific, and we rely on the within-model contrasts that survive correction with both templates pooled. Inference is per model at 48 matters, so effects of a few points are not detectable, and the seven models are one per vendor rather than a sampled population, so capability, family, and training recipe are confounded.

9. Conclusion

Under a citation quota, a date restriction, and a sanctions warning, citation integrity degrades from the smallest model upward, with nonexistent citations at 30B, misquotation through the middle of the range, and paraphrase in quotation marks, misattribution, and wrong pages at the top, and three refusals in 1,344 drafts. A checker built from public data detects each class, including the page-level errors thought to require commercial databases, and retrieval nearly removes the existence failures while leaving paraphrase in quotation marks in place. Returned to the model, the checker's findings repair citation existence; flagged quotations are removed far more often than corrected, and false flags remove correct ones. We therefore recommend placing verification at the gate and reporting verifier precision for any system that acts on its output.

10. Ethics and Release

All source material is public court data; no generated draft was filed. <https://github.com/hardy30894/citation-under-pressure> holds the instruments, drafts, records, and analysis ledger.

References

- [1] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); 2023. Opinion and order on sanctions, No. 1:22-cv-01461 (PKC), June 22, 2023.
- [2] Charlotin D. AI Hallucination Cases Database; 2025. Accessed 2 September 2026; 2,008 cases listed. <https://www.damiencharlotin.com/hallucinations/>.
- [3] Liu P, Stambach D, Henderson P. Who Checks the Citations? Benchmarking Legal Hallucination Detection; 2026. Accepted at the Conference on Language Modeling (COLM) 2026. Available from: <https://arxiv.org/abs/2606.21155>. doi:10.48550/arXiv.2606.21155.
- [4] Chen S, Yin H, Zhou S. LegalCiteBench: Evaluating Citation Reliability in Legal Language Models; 2026. Available from: <https://arxiv.org/abs/2605.10186>. doi:10.48550/arXiv.2605.10186.
- [5] Zhao C, Tang Y, Qian Y. Do Deployment Constraints Make LLMs Hallucinate Citations? An Empirical Study across Four Models and Five Prompting Regimes; 2026. Available from: <https://arxiv.org/abs/2603.07287>. doi:10.48550/arXiv.2603.07287.
- [6] Harvard Law School Library Innovation Lab. Caselaw Access Project; 2024. Digitized collection of official, book-published U.S. case law; accessed September 2026. <https://case.law/>.
- [7] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*. 2024;16(1):64-93. doi:10.1093/jla/laae003.
- [8] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*. 2025;22(2):216-42. doi:10.1111/jels.12413.
- [9] Gehring J, Zheng K, Copet J, Mella V, Cohen T, Synnaeve G. RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning. In: *Proceedings of the 42nd International Conference on Machine Learning*. vol. 267 of *Proceedings of Machine Learning Research*. PMLR; 2025. p. 19034-55. Available from: <https://proceedings.mlr.press/v267/gehring25a.html>.
- [10] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Chicago, IL, USA: ACM; 2025. p. 425-9. doi:10.1145/3769126.3769245.
- [11] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 86-97. doi:10.3233/FAIA251579.
- [12] Sanchi M, Novotná T. Towards Hybrid Evaluation Methodologies for Large Language Models in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 389-92. doi:10.3233/FAIA241279.
- [13] Garneau N, Olsen HP, Corduant A, Tarissan F. Combining Network and Text to Provide Legal Pincites. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 282-7. doi:10.3233/FAIA241256.
- [14] Zhang L, Savelka J, Ashley KD. Do LLMs Truly “Understand” when a Precedent Is Overruled? In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 239-50. doi:10.3233/FAIA251592.
- [15] Nguyen HT, Satoh K. ConsRAG: Minimize LLM Hallucinations in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 327-32. doi:10.3233/FAIA241263.
- [16] Trozze A, Davies T, Kleinberg B. Large language models in cryptocurrency securities cases: can a GPT model meaningfully assist lawyers? *Artificial Intelligence and Law*. 2025;33(3):691-737. doi:10.1007/s10506-024-09399-6.
- [17] Harasta J, Novotná T, Savelka J. It cannot be right if it was written by AI: on lawyers’ preferences of documents perceived as authored by an LLM vs a human. *Artificial Intelligence and Law*. 2026;34(1):153-90. doi:10.1007/s10506-024-09422-w.
- [18] Gray MA, Savelka J, Oliver WM, Ashley KD. Generating Legal Arguments with Automatically Identified Factor Magnitudes. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 199-208. doi:10.1145/3769126.3769212.
- [19] Pires R, Malaquias Junior R, Nogueira R. Automatic Legal Writing Evaluation of LLMs. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 420-4. doi:10.1145/3769126.3769227.

- [20] Khatri M, Sheik R, Wadhwa P, Satija G, Kumar Y, Shah RR, et al. CiteCaseLAW: Citation Worthiness Detection in Caselaw for Legal Assistive Writing. In: Legal Knowledge and Information Systems: JURIX 2023, vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 257-62. doi:10.3233/FAIA230971.
- [21] Sadeghian A, Sundaram L, Wang DZ, Hamilton WF, Branting K, Pfeifer C. Automatic semantic edge labeling over legal citation graphs. Artificial Intelligence and Law. 2018;26(2):127-44. doi:10.1007/s10506-018-9217-1.
- [22] Dadgostari F, Guim M, Beling PA, Livermore MA, Rockmore DN. Modeling law search as prediction. Artificial Intelligence and Law. 2021;29(1):3-34. doi:10.1007/s10506-020-09261-5.
- [23] Mumford J, Florimonte F, Atkinson K, Dzehtsiarou K. Context-Aware Citation Networks: A Human-AI Dataset, Analysis, and Tool. In: Legal Knowledge and Information Systems: JURIX 2025, vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 145-55. doi:10.3233/FAIA251584.
- [24] Spaeth HJ, Epstein L, Nelson MJ, Martin AD, Segal JA, Ruger TJ, et al.. Supreme Court Database, Version 2025 Release 01; 2025. Modern database, terms 1946 to 2024. <https://scdb.la.psu.edu/>.
- [25] Ortega JE, Joshi DD, Borkowski MP. Large-Language Memorization During the Classification of United States Supreme Court Cases; 2025. Available from: <https://arxiv.org/abs/2512.13654>. doi:10.48550/arXiv.2512.13654.
- [26] Federal Rules of Civil Procedure, Rule 1 (Scope and Purpose); 2024. "These rules govern the procedure in all civil actions and proceedings in the United States district courts". https://www.law.cornell.edu/rules/frcp/rule_1.
- [27] Cushman J, Dahl M, Lissner M. eyecite: A tool for parsing legal citations. Journal of Open Source Software. 2021;6(66):3617. Available from: <https://doi.org/10.21105/joss.03617>. doi:10.21105/joss.03617.
- [28] Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. Biometrika. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
- [29] Norman JD, Rivera MU, Hughes DA. Reliability without Validity: A Systematic, Large-Scale Evaluation of LLM-as-a-Judge Models Across Agreement, Consistency, and Bias; 2026. Available from: <https://arxiv.org/abs/2606.19544>. doi:10.48550/arXiv.2606.19544.
- [30] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al.. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena; 2023. NeurIPS 2023 Datasets and Benchmarks Track. Available from: <https://arxiv.org/abs/2306.05685>. doi:10.48550/arXiv.2306.05685.